

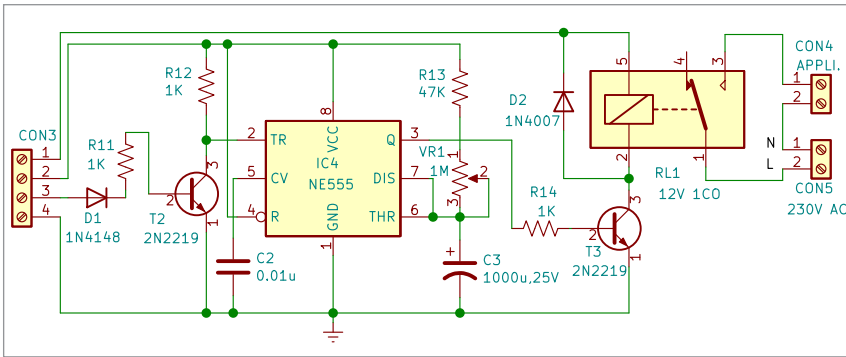


Fig. 5: Circuit diagram of appliance control

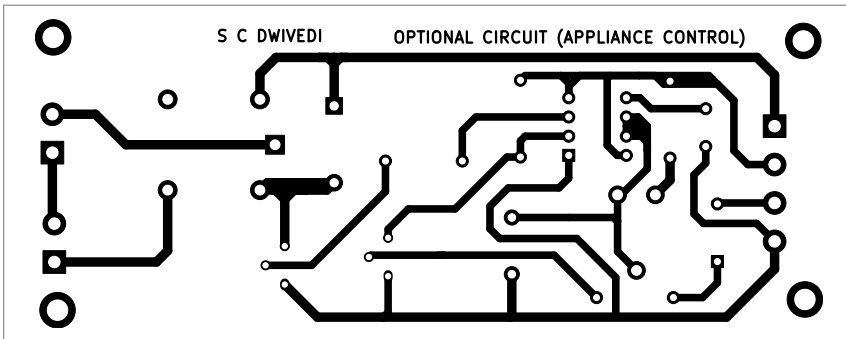


Fig. 6: Actual-size PCB of appliance control circuit

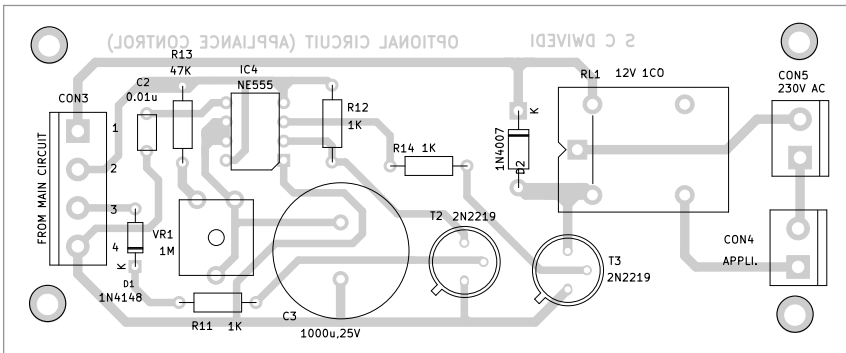


Fig. 7: Component layout of the PCB

4. After assembling the circuit on PCB, enclose it in a suitable box. Fix LED1 and buzzer in front of the box. Place the switches (S1 through S8) through twin flexible wires in different room numbers (RN1 through RN8) for operating the buzzer.

After assembling the components, install the unit near the common room in such a way that when somebody from any room number (RN1 through RN8) presses

any switch, the buzzer sounds there to draw attention of the concerned person(s). CON2 is used for the optional appliance control.

Optional circuit (appliance control)

The circuit diagram for appliance control is shown in Fig. 5. It is built around timer NE555 (IC4), two transistors 2N2219 (T2 and T3), 12V one changeover (SPDT) relay, and a

TABLE 2
APPLIANCE-ON TIMING

Value of R = (R13+VR1)	Value of C = C3 = 1000μF, 25V	Appliance on
55k set via VR1	1000μF	1 minute
272.7k set via VR1	1000μF	5 minutes
545.4k set via VR1	1000μF	10 minutes
818.2k set via VR1	1000μF	15 minutes

PARTS LIST: MULTISWITCH CONTROL BUZZER/APPLIANCES

Semiconductors:

- IC1 - LM7805, 5V voltage regulator
- IC2 (IC2A-IC2E), IC3 (IC3A-IC3E) - 74LS86 quad EX-OR gate
- T1 - 2N2219 NPN transistor
- BR1 - 1A bridge rectifier
- LED1 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

- R1-R10 - 1-kilo-ohm

Capacitors:

- C1 - 1000μF, 35V electrolytic

Miscellaneous:

- CON1 - 2-pin connector
- CON2 - 4-pin male connector
- S1-S8 - Push-to-on switch
- PZ1 - Buzzer
- X1 - 230V AC primary to 15V, 500mA secondary transformer

PARTS LIST: OPTIONAL (APPLIANCE CONTROL)

Semiconductors:

- IC4 - NE555 timer
- T2, T3 - 2N2219 NPN transistor
- D1 - 1N4148 signal timer
- D2 - 1N4007 rectifier diode

Resistors (all 1/4-watt, ±5% carbon):

- R11, R12, R14 - 1-kilo-ohm
- R13 - 47-kilo-ohm
- VR1 - 1-mega-ohm potentiometer

Capacitors:

- C2 - 1000μF, 35V electrolytic
- C3 - 1000μF, 25V electrolytic

Miscellaneous:

- CON3 - 2-pin female connector
- CON4 - 2-pin connector
- CON5 - 2-pin connector
- RL1 - 12V, 1 changeover (1CO) relay
- 230V AC appliance

few other components. This circuit is controlled by the buzzer circuit by interconnecting CON2 and CON3 and removing the buzzer.

The working of this circuit is